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Ответы 67

Задача 1

1. Укажите решение неравенства $-3 - x \geq x - 6$.

   1) $(-\infty; 1{,}5]$

   2) $[1{,}5; +\infty)$

   3) $(-\infty; 4{,}5]$

   4) $[4{,}5; +\infty)$

2. Укажите решение неравенства $3 - 2x \geq 8x - 1$.

   1) $[-0{,}2; +\infty)$

   2) $(-\infty; 0{,}4]$

   3) $[0{,}4; +\infty)$

   4) $(-\infty; -0{,}2]$

3. Укажите решение неравенства $-3 - x > 4x + 7$.

   1) $(-\infty; -0{,}8)$

   2) $(-\infty; -2)$

   3) $(-2; +\infty)$

   4) $(-0{,}8; +\infty)$

4. Укажите решение неравенства $-9 - 6x > 9x + 9$.

   1) $(-\infty; -1{,}2)$

   2) $(0; +\infty)$

   3) $(-1{,}2; +\infty)$

   4) $(-\infty; 0)$

5. Укажите решение неравенства $-3 - 3x > 7x - 9$.

   1) $(0{,}6; +\infty)$

   2) $(-\infty; 1{,}2)$

   3) $(1{,}2; +\infty)$

   4) $(-\infty; 0{,}6)$

6. Укажите решение неравенства $6 - 7x \leq 3x - 7$.

   1) $[0{,}1; +\infty)$
   2) $(-\infty; 1{,}3]$
   3) $[1{,}3; +\infty)$
   4) $(-\infty; 0{,}1]$

7. Укажите решение неравенства $-3 - 5x \leq x + 3$.

   1) $(-\infty; 0]$
   2) $[-1; +\infty)$
   3) $[0; +\infty)$
   4) $(-\infty; -1]$

8. Укажите решение неравенства $-3 - x < 4x + 7$.

   1) $(-\infty; -0{,}8)$
   2) $(-2; +\infty)$
   3) $(-\infty; -2)$
   4) $(-0{,}8; +\infty)$

9. Укажите решение неравенства $-9 - 6x < 9x + 9$.

   1) $(-\infty; -1{,}2)$
   2) $(-1{,}2; +\infty)$
   3) $(0; +\infty)$
   4) $(-\infty; 0)$

10. Укажите решение неравенства $-3 - 3x < 7x - 9$.

   1) $(-\infty; 0{,}6)$
   2) $(-\infty; 1{,}2)$
   3) $(0{,}6; +\infty)$
   4) $(1{,}2; +\infty)$

Задача 2

1. Укажите решение неравенства $4x - 4 \geq 9x + 6$.

   1) $[-0{,}4; +\infty)$

   2) $(-\infty; -2]$

   3) $[-2; +\infty)$

   4) $(-\infty; -0{,}4]$

2. Укажите решение неравенства $2x - 8 \geq 4x + 6$.

   1) $(-\infty; -7]$

   2) $(-\infty; 1]$

   3) $[1; +\infty)$

   4) $[-7; +\infty)$

3. Укажите решение неравенства $5x + 4 \leq x + 6$.

   1) $(-\infty; 0{,}5]$

   2) $(-\infty; 2{,}5]$

   3) $[0{,}5; +\infty)$

   4) $[2{,}5; +\infty)$

4. Укажите решение неравенства $4x - 4 \leq 9x + 6$.

   1) $(-\infty; -0{,}4]$

   2) $[-2; +\infty)$

   3) $(-\infty; -2]$

   4) $[-0{,}4; +\infty)$

5. Укажите решение неравенства $2x - 8 \leq 4x + 6$.

   1) $[-7; +\infty)$

   2) $(-\infty; -7]$

   3) $[1; +\infty)$

   4) $(-\infty; 1]$

6. Укажите решение неравенства $8x - 8 > 7x + 6$.

   1) $(-\infty; 14)$

   2) $(14; +\infty)$

   3) $(-2; +\infty)$

   4) $(-\infty; -2)$

7. Укажите решение неравенства $2x - 8 > 4x + 6$.

   1) $(-\infty; 1)$

   2) $(1; +\infty)$

   3) $(-\infty; -7)$

   4) $(-7; +\infty)$

8. Укажите решение неравенства $5x + 4 < x + 6$.

   1) $(-\infty; 0{,}5)$

   2) $(2{,}5; +\infty)$

   3) $(-\infty; 2{,}5)$

   4) $(0{,}5; +\infty)$

9. Укажите решение неравенства $7x + 9 < 9x - 8$.

   1) $(-0{,}5; +\infty)$

   2) $(8{,}5; +\infty)$

   3) $(-\infty; 8{,}5)$

   4) $(-\infty; -0{,}5)$

10. Укажите решение неравенства $2x - 4 < 7x - 1$.

    1) $(1; +\infty)$

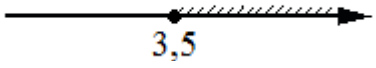
    2) $(-0{,}6; +\infty)$

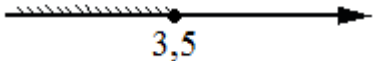
    3) $(-\infty; 1)$

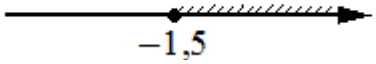
    4) $(-\infty; -0{,}6)$

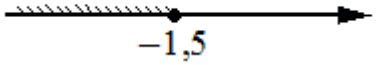
Задача 3

1. Укажите решение неравенства $4x + 5 \geq 6x - 2$.

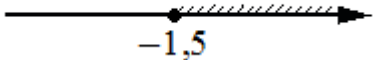
1) 

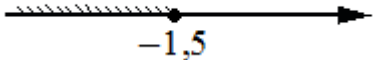
2) 

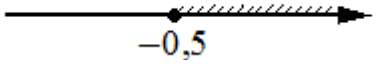
3) 

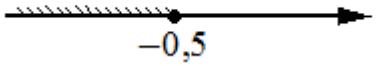
4) 

2. Укажите решение неравенства $x - 1 \leq 3x + 2$.

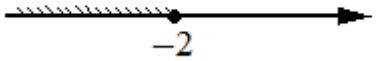
1) 

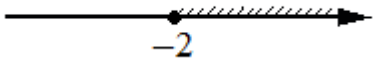
2) 

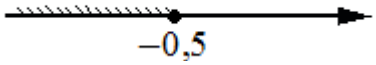
3) 

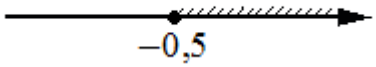
4) 

3. Укажите решение неравенства $3 - x \geq 3x + 5$.

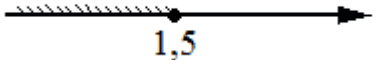
1) 

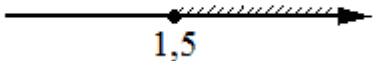
2) 

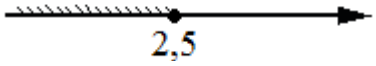
3) 

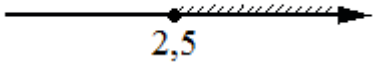
4) 

4. Укажите решение неравенства $2 + x \leq 5x - 8$.

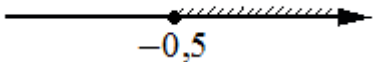
1) 

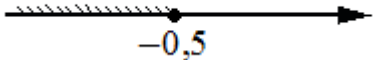
2) 

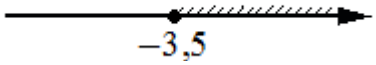
3) 

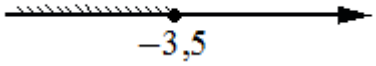
4) 

5. Укажите решение неравенства $4x - 2 \geq -2x - 5$.

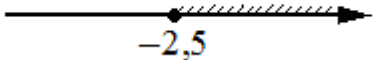
1) 

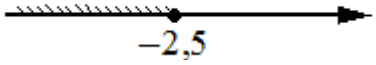
2) 

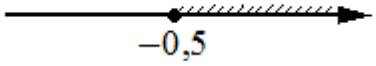
3) 

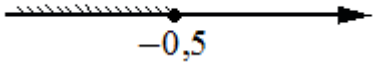
4) 

6. Укажите решение неравенства $2x + 4 \leq -4x + 1$.

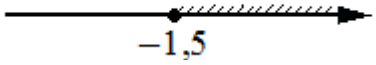
1) 

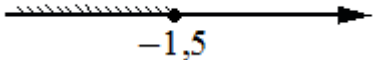
2) 

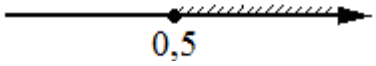
3) 

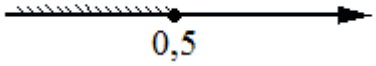
4) 

7. Укажите решение неравенства $4x - 5 \geq 2x - 4$.

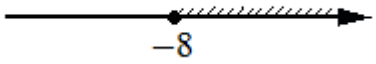
1) 

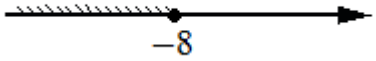
2) 

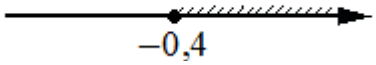
3) 

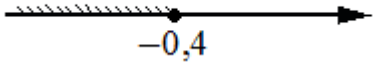
4) 

8. Укажите решение неравенства $-2x + 5 \leq -3x - 3$.

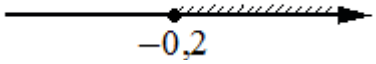
1) 

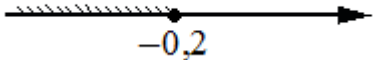
2) 

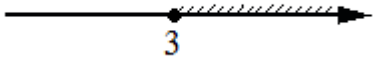
3) 

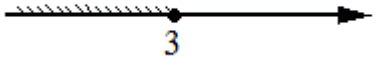
4) 

9. Укажите решение неравенства $x + 4 \geq 4x - 5$.

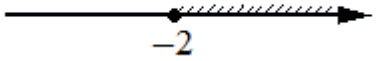
1) 

2) 

3) 

4) 

10. Укажите решение неравенства $x - 2 \leq 4x + 4$.

1)


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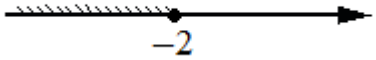

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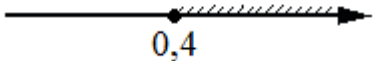

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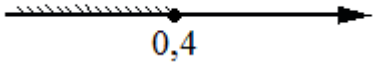


Задача 4

1. Укажите решение неравенства $3x - 2(x - 5) \leq -6$.

   1) $[4; +\infty)$

   2) $(-\infty; 4]$

   3) $(-\infty; -16]$

   4) $[-16; +\infty)$

2. Укажите решение неравенства $2x - 3(x - 7) \leq 3$.

   1) $(-\infty; -24]$

   2) $(-\infty; 18]$

   3) $[18; +\infty)$

   4) $[-24; +\infty)$

3. Укажите решение неравенства $9x - 4(x - 7) \geq -3$.

   1) $[5; +\infty)$

   2) $(-\infty; -6{,}2]$

   3) $[-6{,}2; +\infty)$

   4) $(-\infty; 5]$

4. Укажите решение неравенства $8x - 3(x + 9) \geq -9$.

   1) $[3{,}6; +\infty)$

   2) $[-7{,}2; +\infty)$

   3) $(-\infty; 3{,}6]$

   4) $(-\infty; -7{,}2]$

5. Укажите решение неравенства $2x - 4(3x + 9) \geq -3$.

   1) $(-\infty; -3{,}3]$

   2) $[-3{,}3; +\infty)$

   3) $[3{,}9; +\infty)$

   4) $(-\infty; 3{,}9]$

6. Укажите решение неравенства $8x - 3(3x + 8) \geq 9$.

   1) $[15; +\infty)$

   2) $(-\infty; -33]$

   3) $(-\infty; 15]$

   4) $[-33; +\infty)$

7. Укажите решение неравенства $5x - 2(2x - 8) < -5$.

   1) $(-\infty; 11)$

   2) $(11; +\infty)$

   3) $(-\infty; -21)$

   4) $(-21; +\infty)$

8. Укажите решение неравенства $5x - 3(5x - 8) < -7$.

   1) $(-\infty; 3{,}1)$

   2) $(-1{,}7; +\infty)$

   3) $(-\infty; -1{,}7)$

   4) $(3{,}1; +\infty)$

9. Укажите решение неравенства $3x - 2(x - 2) > -4$.

   1) $(0; +\infty)$

   2) $(-8; +\infty)$

   3) $(-\infty; 0)$

   4) $(-\infty; -8)$

10. Укажите решение неравенства $6x - 3(4x + 1) > 6$.

    1) $(-1{,}5; +\infty)$

    2) $(-\infty; -1{,}5)$

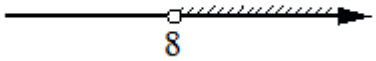
    3) $(-\infty; -0{,}5)$

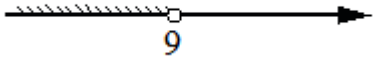
    4) $(-0{,}5; +\infty)$

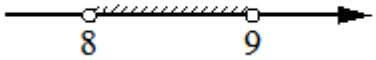
Задача 5

1. Укажите решение системы неравенств

$$\begin{cases} x > 8, \\ 9 - x > 0. \end{cases}$$

1) 

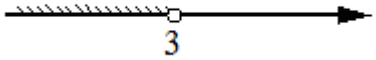
2) 

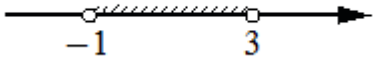
3) 

4) Нет решений

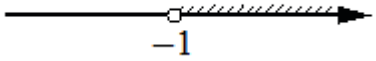
2. Укажите решение системы неравенств

$$\begin{cases} x > -1, \\ 3 - x > 0. \end{cases}$$

1) 

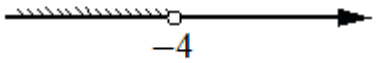
2) 

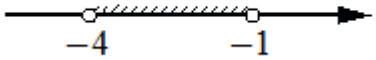
3) Нет решений

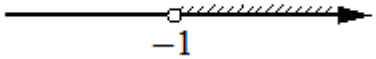
4) 

3. Укажите решение системы неравенств

$$\begin{cases} x > -1, \\ -4 - x > 0. \end{cases}$$

1) 

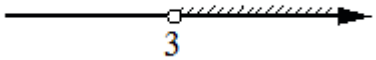
2) 

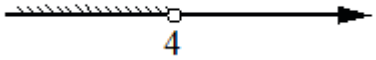
3) 

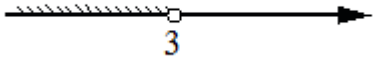
4) Нет решений

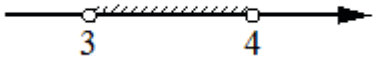
4. Укажите решение системы неравенств

$$\begin{cases} x < 3, \\ 4 - x > 0. \end{cases}$$

1) 

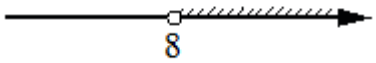
2) 

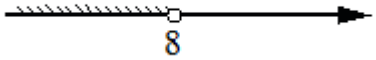
3) 

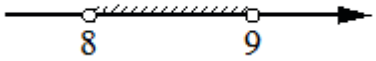
4) 

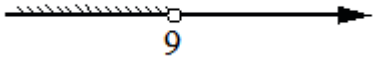
5. Укажите решение системы неравенств

$$\begin{cases} x < 9, \\ 8 - x > 0. \end{cases}$$

1) 

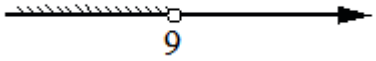
2) 

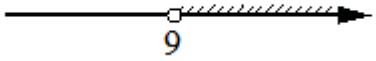
3) 

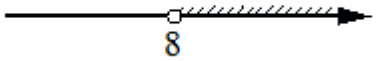
4) 

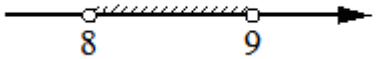
6. Укажите решение системы неравенств

$$\begin{cases} x > 8, \\ 9 - x < 0. \end{cases}$$

1) 

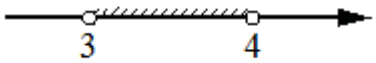
2) 

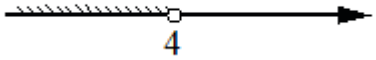
3) 

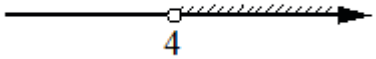
4) 

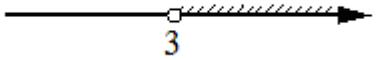
7. Укажите решение системы неравенств

$$\begin{cases} x > 3, \\ 4 - x < 0. \end{cases}$$

1) 

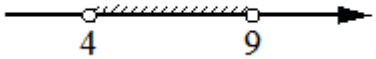
2) 

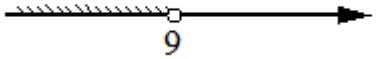
3) 

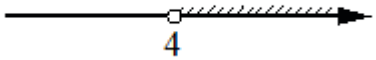
4) 

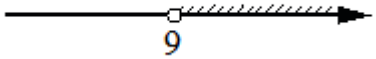
8. Укажите решение системы неравенств

$$\begin{cases} x > 9, \\ 4 - x < 0. \end{cases}$$

1) 

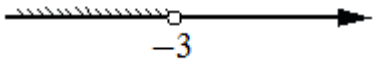
2) 

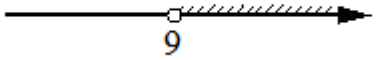
3) 

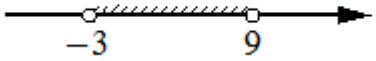
4) 

9. Укажите решение системы неравенств

$$\begin{cases} x < -3, \\ 9 - x < 0. \end{cases}$$

1) 

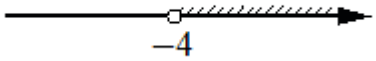
2) 

3) 

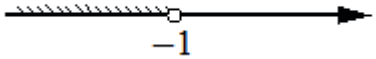
4) Нет решений

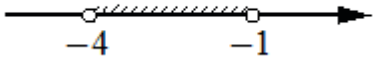
10. Укажите решение системы неравенств

$$\begin{cases} x < -1, \\ -4 - x < 0. \end{cases}$$

1) 

2) Нет решений

3) 

4) 

Задача 6

1. Укажите решение системы неравенств

$$\begin{cases} x - 2{,}6 \leq 0 \\ x - 1 \geq 1 \end{cases}$$

1) $[2; 2{,}6]$
2) $(-\infty; 2{,}6]$
3) $(-\infty; 2] \cup [2{,}6; +\infty)$
4) $[2; +\infty)$

2. Укажите решение системы неравенств

$$\begin{cases} x - 5{,}2 \geq 0 \\ x + 4 \leq 10 \end{cases}$$

1) $(-\infty; 5{,}2] \cup [6; +\infty)$
2) $[5{,}2; +\infty)$
3) $[6; +\infty)$
4) $[5{,}2; 6]$

3. Укажите решение системы неравенств

$$\begin{cases} x - 3 \geq 0 \\ x - 0{,}2 \geq 2 \end{cases}$$

1) $[2{,}2; +\infty)$
2) $[3; +\infty)$
3) $[2{,}2; 3]$
4) $(-\infty; 2{,}2] \cup [3; +\infty)$

4. Укажите решение системы неравенств

$$\begin{cases} x + 2{,}8 \leq 0 \\ x + 0{,}3 \leq -1{,}4 \end{cases}$$

1) $(-\infty; -2{,}8]$
2) $(-\infty; -2{,}8] \cup [-2{,}7; +\infty)$
3) $[-2{,}8; -2{,}7]$
4) $[-2{,}7; +\infty)$

5. Укажите решение системы неравенств

$$\begin{cases} x + 3{,}4 \leq 0 \\ x + 5 \geq 1 \end{cases}$$

1) $(-\infty; -4]$
2) $[-3{,}4; +\infty)$
3) $[-4; -3{,}4]$
4) $(-\infty; -4] \cup [-3{,}4; +\infty)$

6. Укажите решение системы неравенств

$$\begin{cases} x + 4 \geq -3{,}4 \\ x + 5 \leq 0 \end{cases}$$

1) $[-7{,}4; -5]$
2) $[-5; +\infty)$
3) $(-\infty; -7{,}4]$
4) $(-\infty; -7{,}4] \cup [-5; +\infty)$

7. Укажите решение системы неравенств

$$\begin{cases} x + 4 \geq -1 \\ x + 1{,}4 \geq 0 \end{cases}$$

1) $[-5; +\infty)$
2) $[-1{,}4; +\infty)$
3) $[-5; -1{,}4]$
4) $(-\infty; -5] \cup [-1{,}4; +\infty)$

8. Укажите решение системы неравенств

$$\begin{cases} x + 0{,}6 \leq 0 \\ x - 1 \geq -4 \end{cases}$$

1) $(-\infty; -3]$
2) $[-0{,}6; +\infty)$
3) $(-\infty; -3] \cup [-0{,}6; +\infty)$
4) $[-3; -0{,}6]$

9. Укажите решение системы неравенств

$$\begin{cases} x + 3{,}6 \leq 0 \\ x + 2 \leq -1 \end{cases}$$

1) $(-\infty; -3{,}6] \cup [-3; +\infty)$
2) $(-\infty; -3{,}6]$
3) $[-3{,}6; -3]$
4) $[-3{,}6; +\infty)$

10. Укажите решение системы неравенств

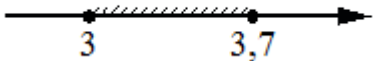
$$\begin{cases} x - 6{,}6 \geq 0 \\ x + 1 \geq 5 \end{cases}$$

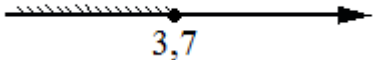
1) $[4; +\infty)$
2) $[4; 6{,}6]$
3) $[6{,}6; +\infty)$
4) $(-\infty; 4]$

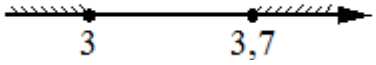
Задача 7

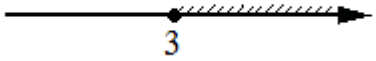
1. Укажите решение системы неравенств

$$\begin{cases} x - 3{,}7 \leq 0 \\ x - 2 \geq 1 \end{cases}$$

1) 

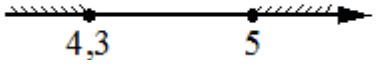
2) 

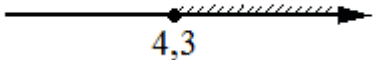
3) 

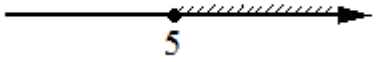
4) 

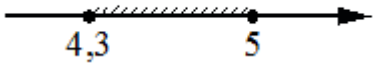
2. Укажите решение системы неравенств

$$\begin{cases} x - 4{,}3 \geq 0 \\ x + 5 \leq 10 \end{cases}$$

1) 

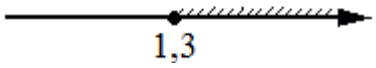
2) 

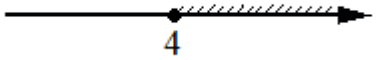
3) 

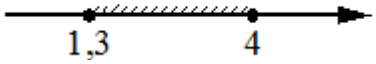
4) 

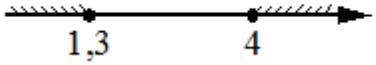
3. Укажите решение системы неравенств

$$\begin{cases} x - 4 \geq 0 \\ x - 0{,}3 \geq 1 \end{cases}$$

1) 

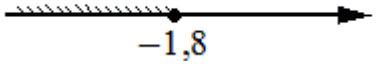
2) 

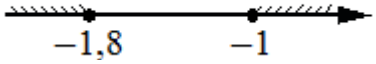
3) 

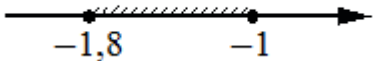
4) 

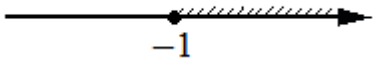
4. Укажите решение системы неравенств

$$\begin{cases} x + 1{,}8 \leq 0 \\ x + 0{,}5 \leq -0{,}5 \end{cases}$$

1) 

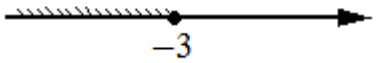
2) 

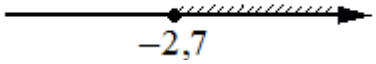
3) 

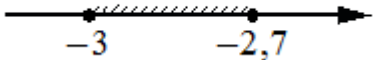
4) 

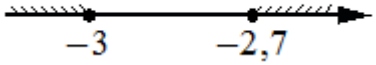
5. Укажите решение системы неравенств

$$\begin{cases} x + 2{,}7 \leq 0 \\ x + 4 \geq 1 \end{cases}$$

1) 

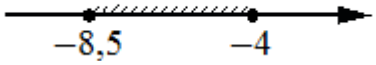
2) 

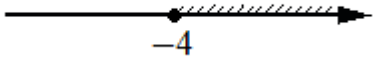
3) 

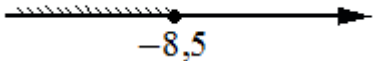
4) 

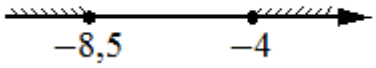
6. Укажите решение системы неравенств

$$\begin{cases} x + 4 \geq -4{,}5 \\ x + 4 \leq 0 \end{cases}$$

1) 

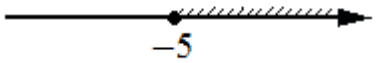
2) 

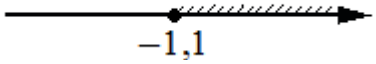
3) 

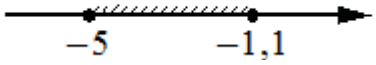
4) 

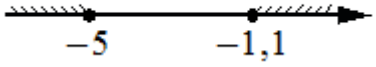
7. Укажите решение системы неравенств

$$\begin{cases} x+3 \geq -2 \\ x+1{,}1 \geq 0 \end{cases}$$

1) 

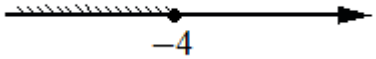
2) 

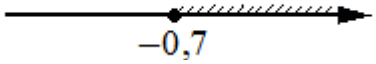
3) 

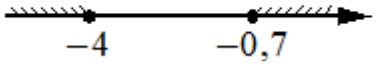
4) 

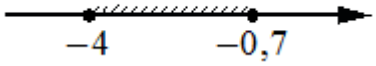
8. Укажите решение системы неравенств

$$\begin{cases} x+0{,}7 \leq 0 \\ x-1 \geq -5 \end{cases}$$

1) 

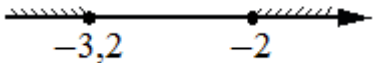
2) 

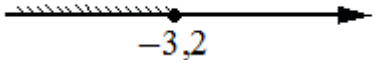
3) 

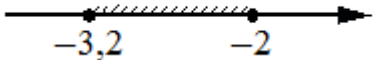
4) 

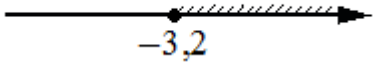
9. Укажите решение системы неравенств

$$\begin{cases} x + 3{,}2 \leq 0 \\ x + 1 \leq -1 \end{cases}$$

1) 

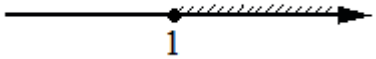
2) 

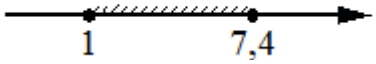
3) 

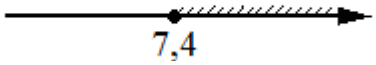
4) 

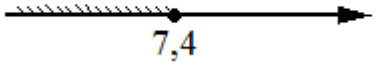
10. Укажите решение системы неравенств

$$\begin{cases} x - 7{,}4 \geq 0 \\ x + 2 \geq 3 \end{cases}$$

1) 

2) 

3) 

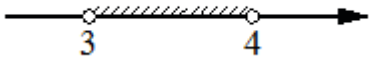
4) 

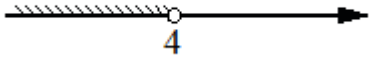
Задача 8

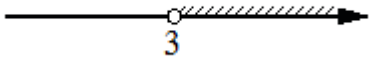
1. Укажите решение системы неравенств

$$\begin{cases} -9 + 3x > 0 \\ 2 - 3x > -10 \end{cases}$$

1) Нет решений

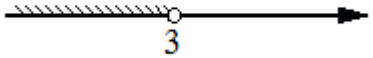
2) 

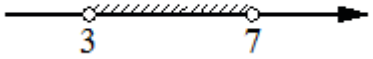
3) 

4) 

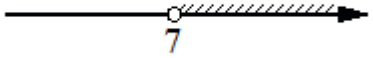
2. Укажите решение системы неравенств

$$\begin{cases} -35 + 5x > 0 \\ 6 - 3x > -3 \end{cases}$$

1) 

2) 

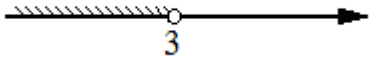
3) Нет решений

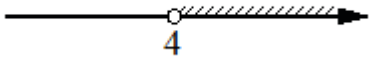
4) 

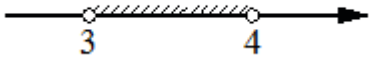
3. Укажите решение системы неравенств

$$\begin{cases} -12 + 3x > 0 \\ 9 - 4x > -3 \end{cases}$$

1) Нет решений

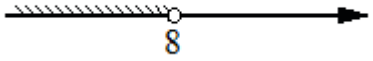
2) 

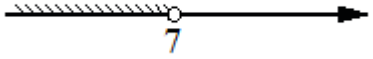
3) 

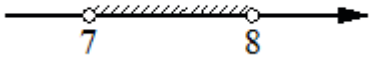
4) 

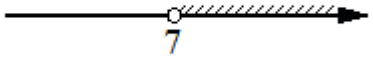
4. Укажите решение системы неравенств

$$\begin{cases} -35 + 5x < 0 \\ 6 - 3x > -18 \end{cases}$$

1) 

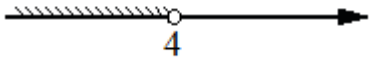
2) 

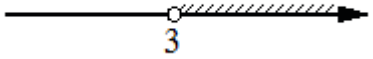
3) 

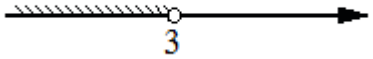
4) 

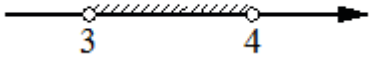
5. Укажите решение системы неравенств

$$\begin{cases} -9+3x<0 \\ 2-3x>-10 \end{cases}$$

1) 

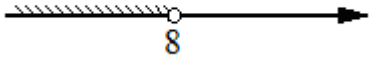
2) 

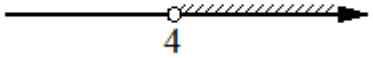
3) 

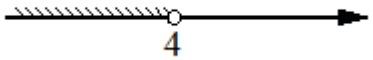
4) 

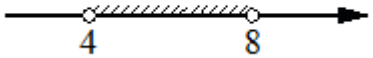
6. Укажите решение системы неравенств

$$\begin{cases} -12+3x<0 \\ 9-4x>-23 \end{cases}$$

1) 

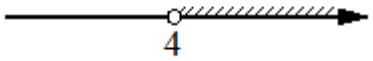
2) 

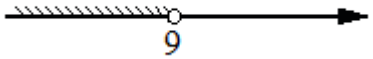
3) 

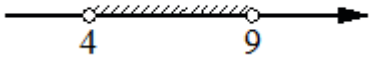
4) 

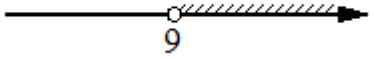
7. Укажите решение системы неравенств

$$\begin{cases} -27 + 3x > 0 \\ 6 - 3x < -6 \end{cases}$$

1) 

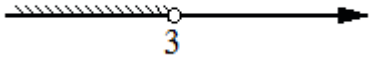
2) 

3) 

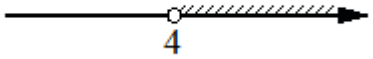
4) 

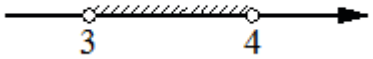
8. Укажите решение системы неравенств

$$\begin{cases} -9 + 3x < 0 \\ 2 - 3x < -10 \end{cases}$$

1) 

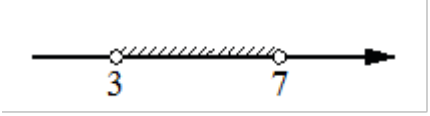
2) Нет решений

3) 

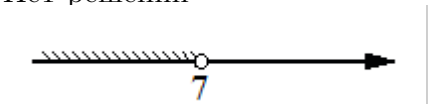
4) 

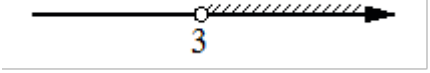
9. Укажите решение системы неравенств

$$\begin{cases} -35 + 5x < 0 \\ 6 - 3x < -3 \end{cases}$$

1) 

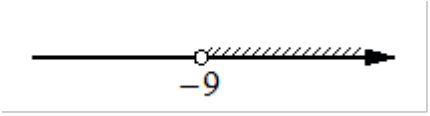
2) Нет решений

3) 

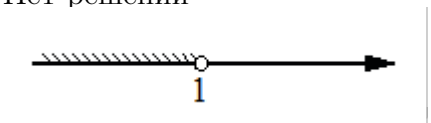
4) 

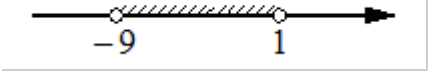
10. Укажите решение системы неравенств

$$\begin{cases} -5 + 5x < 0 \\ 4 - 3x < 31 \end{cases}$$

1) 

2) Нет решений

3) 

4) 

Задача 9

1. Укажите решение системы неравенств

$$\begin{cases} -8 + 4x > 0 \\ 4 - 3x > -8 \end{cases}$$

1) Нет решений
2) $(-\infty; 4)$
3) $(2; +\infty)$
4) $(2; 4)$

2. Укажите решение системы неравенств

$$\begin{cases} -48 + 6x > 0 \\ 6 - 5x > -4 \end{cases}$$

1) $(2; 8)$
2) $(-\infty; 2)$
3) Нет решений
4) $(8; +\infty)$

3. Укажите решение системы неравенств

$$\begin{cases} -10 + 2x > 0 \\ 7 - 6x > -5 \end{cases}$$

1) Нет решений
2) $(5; +\infty)$
3) $(2; 5)$
4) $(-\infty; 2)$

4. Укажите решение системы неравенств

$$\begin{cases} -35 + 5x < 0 \\ 6 - 3x > -18 \end{cases}$$

1) $(7; 8)$
2) $(-\infty; 7)$
3) $(-\infty; 8)$
4) $(7; +\infty)$

5. Укажите решение системы неравенств

$$\begin{cases} -9+3x<0 \\ 2-3x>-10 \end{cases}$$

1) $(-\infty;3)$
2) $(-\infty;4)$
3) $(3;+\infty)$
4) $(3;4)$

6. Укажите решение системы неравенств

$$\begin{cases} -12+3x<0 \\ 9-4x>-23 \end{cases}$$

1) $(-\infty;8)$
2) $(-\infty;4)$
3) $(4;8)$
4) $(4;+\infty)$

7. Укажите решение системы неравенств

$$\begin{cases} -27 + 3x > 0 \\ 6 - 3x < -6 \end{cases}$$

1) $(4; +\infty)$
2) $(4; 9)$
3) $(9; +\infty)$
4) $(-\infty; 9)$

8. Укажите решение системы неравенств

$$\begin{cases} -12 + 3x < 0 \\ 2 - 7x < -33 \end{cases}$$

1) $(-\infty; 4)$
2) Нет решений
3) $(4; 5)$
4) $(5; +\infty)$

9. Укажите решение системы неравенств

$$\begin{cases} -36 + 4x < 0 \\ 5 - 4x < -3 \end{cases}$$

1) $(2; +\infty)$
2) Нет решений
3) $(-\infty; 9)$
4) $(2; 9)$

10. Укажите решение системы неравенств

$$\begin{cases} -5 + 5x < 0 \\ 4 - 3x < 31 \end{cases}$$

1) $(-9; 1)$
2) Нет решений
3) $(-9; +\infty)$
4) $(-\infty; 1)$

Задача 10

1. Укажите решение неравенства $(x+2)(x-7) \leq 0$.

   1) $[-2; 7]$

   2) $(-\infty; -2] \cup [7; +\infty)$

   3) $(-\infty; 7]$

   4) $(-\infty; -2]$

2. Укажите решение неравенства $(x+3)(x-5) \leq 0$.

   1) $(-\infty; -3]$

   2) $[-3; 5]$

   3) $(-\infty; 5]$

   4) $(-\infty; -3] \cup [5; +\infty)$

3. Укажите решение неравенства $(x+4)(x-8) \leq 0$.

   1) $(-\infty; 8]$

   2) $(-\infty; -4] \cup [8; +\infty)$

   3) $[-4; 8]$

   4) $(-\infty; -4]$

4. Укажите решение неравенства $(x+6)(x-1) < 0$.

   1) $(-\infty; 1)$

   2) $(-\infty; -6)$

   3) $(-\infty; -6) \cup (1; +\infty)$

   4) $(-6; 1)$

5. Укажите решение неравенства $(x+9)(x-4) < 0$.

   1) $(-9; 4)$

   2) $(-\infty; -9) \cup (4; +\infty)$

   3) $(-\infty; -9)$

   4) $(-\infty; 4)$

6. Укажите решение неравенства $(x+2)(x-10)>0$.

   1) $(-2;10)$

   2) $(-\infty;-2)\cup(10;+\infty)$

   3) $(10;+\infty)$

   4) $(-2;+\infty)$

7. Укажите решение неравенства $(x+3)(x-6)>0$.

   1) $(6;+\infty)$

   2) $(-3;+\infty)$

   3) $(-\infty;-3)\cup(6;+\infty)$

   4) $(-3;6)$

8. Укажите решение неравенства $(x+5)(x-9)>0$.

   1) $(-5;+\infty)$

   2) $(-5;9)$

   3) $(9;+\infty)$

   4) $(-\infty;-5)\cup(9;+\infty)$

9. Укажите решение неравенства $(x+1)(x-7)\geq 0$.

   1) $(-\infty;-1]\cup[7;+\infty)$

   2) $[-1;+\infty)$

   3) $[-1;7]$

   4) $[7;+\infty)$

10. Укажите решение неравенства $(x+3)(x-8)\geq 0$.

   1) $[-3;8]$

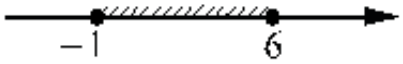
   2) $(-\infty;-3]\cup[8;+\infty)$

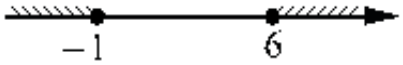
   3) $[8;+\infty)$

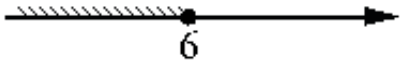
   4) $[-3;+\infty)$

Задача 11

1. Укажите решение неравенства $(x+1)(x-6) \leq 0$.

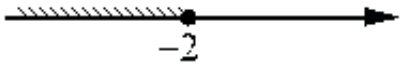
1) 

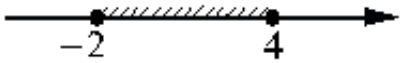
2) 

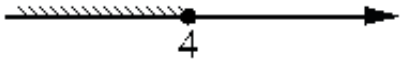
3) 

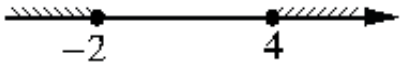
4)

2. Укажите решение неравенства $(x+2)(x-4) \leq 0$.

1) 

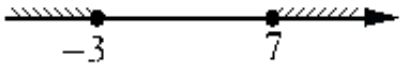
2) 

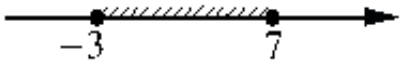
3) 

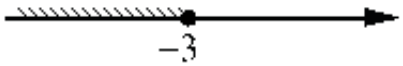
4) 

3. Укажите решение неравенства $(x+3)(x-7) \leq 0$.

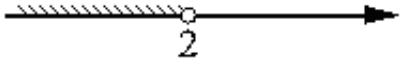
1)

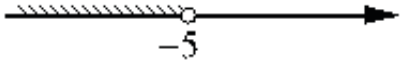
2) 

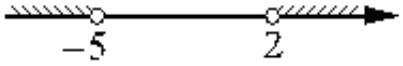
3) 

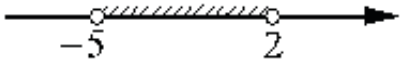
4) 

4. Укажите решение неравенства $(x+5)(x-2)<0$.

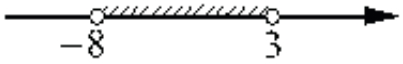
1) 

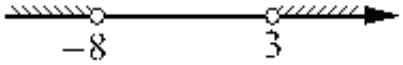
2) 

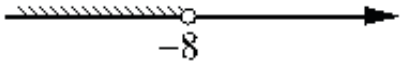
3) 

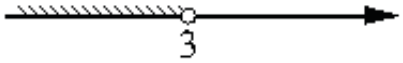
4) 

5. Укажите решение неравенства $(x+8)(x-3)<0$.

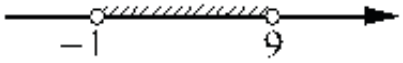
1) 

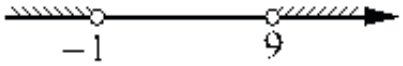
2) 

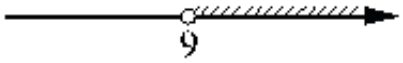
3) 

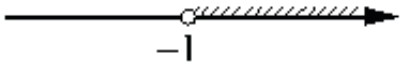
4) 

6. Укажите решение неравенства $(x+1)(x-9)>0$.

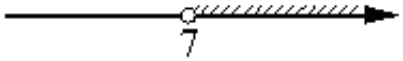
1) 

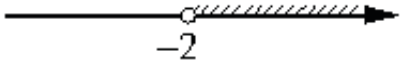
2) 

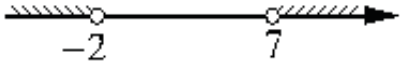
3) 

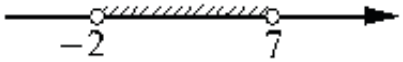
4) 

7. Укажите решение неравенства $(x+2)(x-7)>0$.

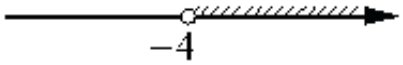
1) 

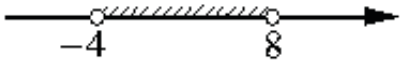
2) 

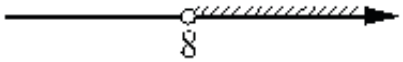
3) 

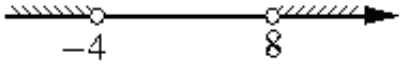
4) 

8. Укажите решение неравенства $(x+4)(x-8)>0$.

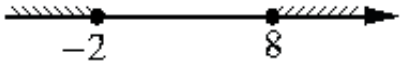
1) 

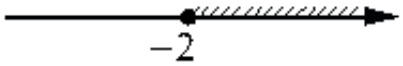
2) 

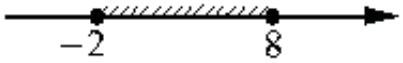
3) 

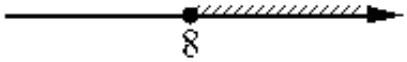
4) 

9. Укажите решение неравенства $(x+2)(x-8)\geq 0$.

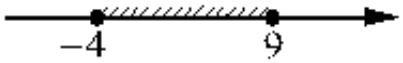
1) 

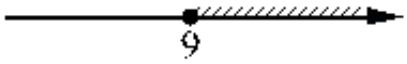
2) 

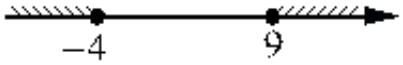
3) 

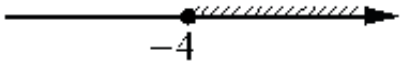
4) 

10. Укажите решение неравенства $(x+4)(x-9) \geq 0$.

1) 

2) 

3) 

4) 

Задача 12

1. Укажите решение неравенства $2x - x^2 \leq 0$.

   1) $(-\infty; 0] \cup [2; +\infty)$

   2) $[0; +\infty)$

   3) $[2; +\infty)$

   4) $[0; 2]$

2. Укажите решение неравенства $10x - x^2 \leq 0$.

   1) $[0; 10]$

   2) $(-\infty; 0] \cup [10; +\infty)$

   3) $[10; +\infty)$

   4) $[0; +\infty)$

3. Укажите решение неравенства $8x - x^2 \leq 0$.

   1) $[8; +\infty)$

   2) $[0; 8]$

   3) $(-\infty; 0] \cup [8; +\infty)$

   4) $[0; +\infty)$

4. Укажите решение неравенства $x - x^2 < 0$.

   1) $(0; 1)$

   2) $(0; +\infty)$

   3) $(1; +\infty)$

   4) $(-\infty; 0) \cup (1; +\infty)$

5. Укажите решение неравенства $4x - x^2 < 0$.

   1) $(-\infty; 0) \cup (4; +\infty)$

   2) $(0; +\infty)$

   3) $(0; 4)$

   4) $(4; +\infty)$

6. Укажите решение неравенства $7x - x^2 \geq 0$.

   1) $[0; +\infty)$

   2) $[7; +\infty)$

   3) $(-\infty; 0] \cup [7; +\infty)$

   4) $[0; 7]$

7. Укажите решение неравенства $6x - x^2 \geq 0$.

   1) $[0; +\infty)$

   2) $(-\infty; 0] \cup [6; +\infty)$

   3) $[0; 6]$

   4) $[6; +\infty)$

8. Укажите решение неравенства $8x - x^2 \geq 0$.

   1) $[0; +\infty)$

   2) $[0; 8]$

   3) $[8; +\infty)$

   4) $(-\infty; 0] \cup [8; +\infty)$

9. Укажите решение неравенства $3x - x^2 > 0$.

   1) $(3; +\infty)$

   2) $(-\infty; 0) \cup (3; +\infty)$

   3) $(0; +\infty)$

   4) $(0; 3)$

10. Укажите решение неравенства $5x - x^2 > 0$.

    1) $(-\infty; 0) \cup (5; +\infty)$

    2) $(0; 5)$

    3) $(5; +\infty)$

    4) $(0; +\infty)$

Задача 13

1. Укажите решение неравенства $x^2 - 49 < 0$.

   1) Нет решений

   2) $(-\infty; +\infty)$

   3) $(-7; 7)$

   4) $(-\infty; -7) \cup (7; +\infty)$

2. Укажите решение неравенства $x^2 - 25 < 0$.

   1) $(-\infty; +\infty)$

   2) Нет решений

   3) $(-5; 5)$

   4) $(-\infty; -5) \cup (5; +\infty)$

3. Укажите решение неравенства $x^2 - 49 > 0$.

   1) $(-7; 7)$

   2) Нет решений

   3) $(-\infty; +\infty)$

   4) $(-\infty; -7) \cup (7; +\infty)$

4. Укажите решение неравенства $x^2 - 36 > 0$.

   1) $(-\infty; +\infty)$

   2) $(-\infty; ; -6) \cup (6; +\infty)$

   3) $(-6; 6)$

   4) Нет решений

5. Укажите решение неравенства $x^2 - 25 > 0$.

   1) $(-\infty; -5) \cup (5; +\infty)$

   2) $(-5; 5)$

   3) Нет решений

   4) $(-\infty; +\infty)$

6. Укажите решение неравенства $x^2 - 64 > 0$.

   1) $(-\infty; +\infty)$

   2) $(-8; 8)$

   3) $(-\infty; -8) \cup (8; +\infty)$

   4) Нет решений

7. Укажите решение неравенства $x^2 - 36 \leq 0$.

   1) $(-\infty; +\infty)$

   2) $(-\infty; -6] \cup [6; +\infty)$

   3) $[-6; 6]$

   4) Нет решений

8. Укажите решение неравенства $x^2 - 49 \geq 0$.

   1) $[-7; 7]$

   2) Нет решений

   3) $(-\infty; -7] \cup [7; +\infty)$

   4) $(-\infty; +\infty)$

9. Укажите решение неравенства $x^2 - 36 \geq 0$.

   1) $(-\infty; +\infty)$

   2) Нет решений

   3) $(-\infty; -6] \cup [6; +\infty)$

   4) $[-6; 6]$

10. Укажите решение неравенства $x^2 - 64 \geq 0$.

    1) $[-8; 8]$

    2) $(-\infty; -8] \cup [8; +\infty)$

    3) Нет решений

    4) $(-\infty; +\infty)$

Задача 14

1. Укажите неравенство, решением которого является любое число.

   1) $x^2 + 70 > 0$

   2) $x^2 - 70 > 0$

   3) $x^2 + 70 < 0$

   4) $x^2 - 70 < 0$

2. Укажите неравенство, решением которого является любое число.

   1) $x^2 - 56 > 0$

   2) $x^2 + 56 > 0$

   3) $x^2 - 56 < 0$

   4) $x^2 + 56 < 0$

3. Укажите неравенство, решением которого является любое число.

   1) $x^2 + 78 > 0$

   2) $x^2 - 78 < 0$

   3) $x^2 + 78 < 0$

   4) $x^2 - 78 > 0$

4. Укажите неравенство, решением которого является любое число.

   1) $x^2 - 15 < 0$

   2) $x^2 + 15 > 0$

   3) $x^2 + 15 < 0$

   4) $x^2 - 15 > 0$

5. Укажите неравенство, решением которого является любое число.

   1) $x^2 - 64 \geq 0$

   2) $x^2 + 64 \leq 0$

   3) $x^2 + 64 \geq 0$

   4) $x^2 - 64 \leq 0$

6. Укажите неравенство, которое **не имеет** решений.

   1) $x^2 - 70 < 0$

   2) $x^2 - 70 > 0$

   3) $x^2 + 70 < 0$

   4) $x^2 + 70 > 0$

7. Укажите неравенство, которое **не имеет** решений.

   1) $x^2 - 56 > 0$

   2) $x^2 + 56 > 0$

   3) $x^2 - 56 < 0$

   4) $x^2 + 56 < 0$

8. Укажите неравенство, которое **не имеет** решений.

   1) $x^2 + 78 > 0$

   2) $x^2 + 78 < 0$

   3) $x^2 - 78 > 0$

   4) $x^2 - 78 < 0$

9. Укажите неравенство, которое **не имеет** решений.

   1) $x^2 + 64 < 0$

   2) $x^2 + 64 > 0$

   3) $x^2 - 64 > 0$

   4) $x^2 - 64 < 0$

10. Укажите неравенство, которое **не имеет** решений.

    1) $x^2 + 15 \geq 0$

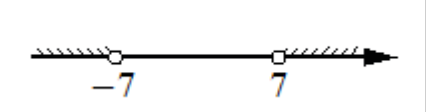
    2) $x^2 - 15 \leq 0$

    3) $x^2 - 15 \geq 0$

    4) $x^2 + 15 \leq 0$

Задача 15

1. Укажите неравенство, решение которого изображено на рисунке.



1) $x^2 - 49 > 0$

2) $x^2 - 49 < 0$

3) $x^2 + 49 < 0$

4) $x^2 + 49 > 0$

2. Укажите неравенство, решение которого изображено на рисунке.

1) $x^2 - 25 > 0$

2) $x^2 - 25 < 0$

3) $x^2 + 25 < 0$

4) $x^2 + 25 > 0$

3. Укажите неравенство, решение которого изображено на рисунке.

1) $x^2 - 36 \leq 0$

2) $x^2 + 36 \geq 0$

3) $x^2 - 36 \geq 0$

4) $x^2 + 36 \leq 0$

4. Укажите неравенство, решение которого изображено на рисунке.

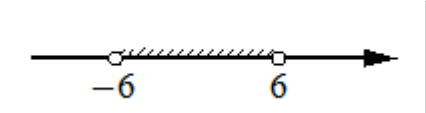
1) $x^2 + 64 \geq 0$

2) $x^2 - 64 \leq 0$

3) $x^2 - 64 \geq 0$

4) $x^2 + 64 \leq 0$

5. Укажите неравенство, решение которого изображено на рисунке.



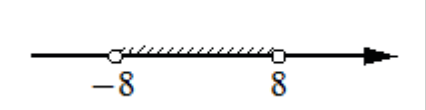
1) $x^2 - 36 > 0$

2) $x^2 + 36 > 0$

3) $x^2 - 36 < 0$

4) $x^2 + 36 < 0$

6. Укажите неравенство, решение которого изображено на рисунке.



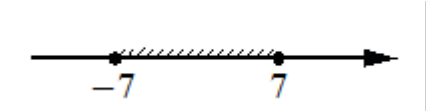
1) $x^2 + 64 > 0$

2) $x^2 - 64 > 0$

3) $x^2 - 64 < 0$

4) $x^2 + 64 < 0$

7. Укажите неравенство, решение которого изображено на рисунке.



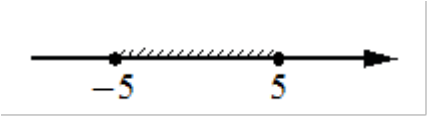
1) $x^2 - 49 \leq 0$

2) $x^2 + 49 \leq 0$

3) $x^2 - 49 \geq 0$

4) $x^2 + 49 \geq 0$

8. Укажите неравенство, решение которого изображено на рисунке.



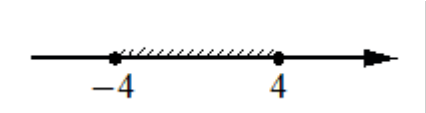
1) $x^2 + 25 \leq 0$

2) $x^2 - 25 \leq 0$

3) $x^2 + 25 \geq 0$

4) $x^2 - 25 \geq 0$

9. Укажите неравенство, решение которого изображено на рисунке.



1) $x^2 + 16 \geq 0$

2) $x^2 - 16 \leq 0$

3) $x^2 + 16 \leq 0$

4) $x^2 - 16 \geq 0$

10. Укажите неравенство, решение которого изображено на рисунке.

1) $x^2 - 9 > 0$

2) $x^2 + 9 > 0$

3) $x^2 - 9 < 0$

4) $x^2 + 9 < 0$

Задача 16

1. Укажите неравенство, которое **не имеет** решений.

   1) $x^2 - 8x - 83 > 0$

   2) $x^2 - 8x + 83 < 0$

   3) $x^2 - 8x - 83 < 0$

   4) $x^2 - 8x + 83 > 0$

2. Укажите неравенство, которое **не имеет** решений.

   1) $x^2 + x + 36 < 0$

   2) $x^2 + x - 36 > 0$

   3) $x^2 + x + 36 > 0$

   4) $x^2 + x - 36 < 0$

3. Укажите неравенство, которое **не имеет** решений.

   1) $x^2 - x + 56 < 0$

   2) $x^2 - x - 56 > 0$

   3) $x^2 - x - 56 < 0$

   4) $x^2 - x + 56 > 0$

4. Укажите неравенство, которое **не имеет** решений.

   1) $x^2 + 6x - 51 > 0$

   2) $x^2 + 6x - 51 < 0$

   3) $x^2 + 6x + 51 > 0$

   4) $x^2 + 6x + 51 < 0$

5. Укажите неравенство, которое **не имеет** решений.

   1) $x^2 + 6x + 12 > 0$

   2) $x^2 + 6x + 12 < 0$

   3) $x^2 + 6x - 12 < 0$

   4) $x^2 + 6x - 12 > 0$

6. Укажите неравенство, которое **не имеет** решений.

   1) $x^2 - 5x + 13 > 0$

   2) $x^2 - 5x - 13 > 0$

   3) $x^2 - 5x - 13 < 0$

   4) $x^2 - 5x + 13 < 0$

7. Укажите неравенство, которое **не имеет** решений.

   1) $x^2 + 9x - 79 < 0$

   2) $x^2 + 9x + 79 > 0$

   3) $x^2 + 9x + 79 < 0$

   4) $x^2 + 9x - 79 > 0$

8. Укажите неравенство, которое **не имеет** решений.

   1) $x^2 - 5x + 53 < 0$

   2) $x^2 - 5x - 53 < 0$

   3) $x^2 - 5x + 53 > 0$

   4) $x^2 - 5x - 53 > 0$

9. Укажите неравенство, которое **не имеет** решений.

   1) $x^2 - 2x - 65 < 0$

   2) $x^2 - 2x - 65 > 0$

   3) $x^2 - 2x + 65 < 0$

   4) $x^2 - 2x + 65 > 0$

10. Укажите неравенство, которое **не имеет** решений.

    1) $x^2 - 3x - 11 < 0$

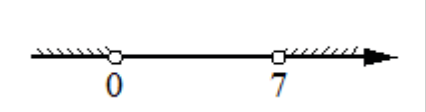
    2) $x^2 - 3x + 11 < 0$

    3) $x^2 - 3x + 11 > 0$

    4) $x^2 - 3x - 11 > 0$

Задача 17

1. Укажите неравенство, решение которого изображено на рисунке.



1) $x^2 - 7x < 0$

2) $x^2 - 49 > 0$

3) $x^2 - 7x > 0$

4) $x^2 - 49 < 0$

2. Укажите неравенство, решение которого изображено на рисунке.

1) $x^2 - 6x < 0$

2) $x^2 - 6x > 0$

3) $x^2 - 36 < 0$

4) $x^2 - 36 > 0$

3. Укажите неравенство, решение которого изображено на рисунке.

1) $x^2 - 64 < 0$

2) $x^2 - 64 > 0$

3) $x^2 - 8x < 0$

4) $x^2 - 8x > 0$

4. Укажите неравенство, решение которого изображено на рисунке.

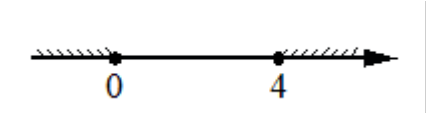
1) $x^2 - 1 \geq 0$

2) $x^2 - x \geq 0$

3) $x^2 - 1 \leq 0$

4) $x^2 - x \leq 0$

5. Укажите неравенство, решение которого изображено на рисунке.



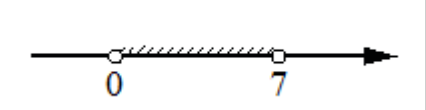
1) $x^2 - 16 \leq 0$

2) $x^2 - 4x \leq 0$

3) $x^2 - 4x \geq 0$

4) $x^2 - 16 \geq 0$

6. Укажите неравенство, решение которого изображено на рисунке.



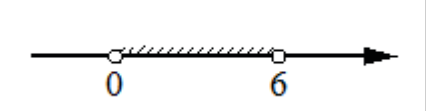
1) $x^2 - 49 < 0$

2) $x^2 - 7x < 0$

3) $x^2 - 49 > 0$

4) $x^2 - 7x > 0$

7. Укажите неравенство, решение которого изображено на рисунке.



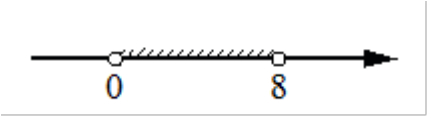
1) $x^2 - 36 < 0$

2) $x^2 - 6x < 0$

3) $x^2 - 6x > 0$

4) $x^2 - 36 > 0$

8. Укажите неравенство, решение которого изображено на рисунке.



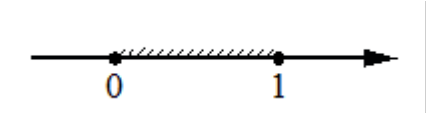
1) $x^2 - 8x < 0$

2) $x^2 - 64 < 0$

3) $x^2 - 8x > 0$

4) $x^2 - 64 > 0$

9. Укажите неравенство, решение которого изображено на рисунке.



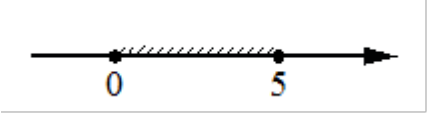
1) $x^2 - 1 \leq 0$

2) $x^2 - x \geq 0$

3) $x^2 - 1 \geq 0$

4) $x^2 - x \leq 0$

10. Укажите неравенство, решение которого изображено на рисунке.



1) $x^2 - 5x \leq 0$

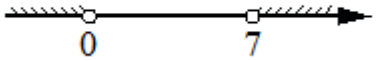
2) $x^2 - 25 \leq 0$

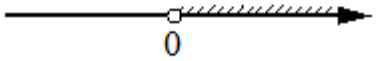
3) $x^2 - 5x \geq 0$

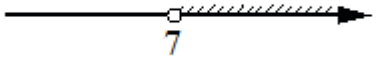
4) $x^2 - 25 \geq 0$

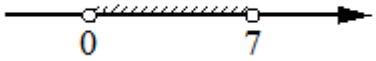
Задача 18

1. Укажите решение неравенства $7x - x^2 < 0$.

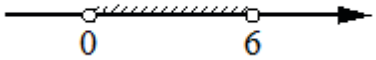
1) 

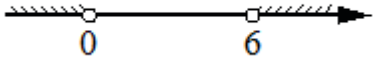
2) 

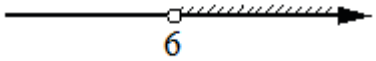
3) 

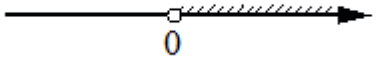
4) 

2. Укажите решение неравенства $6x - x^2 < 0$.

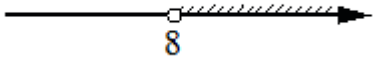
1) 

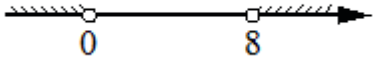
2) 

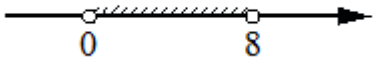
3) 

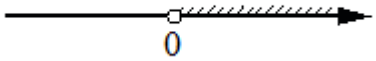
4) 

3. Укажите решение неравенства $8x - x^2 < 0$.

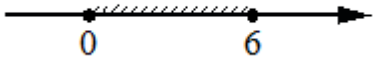
1) 

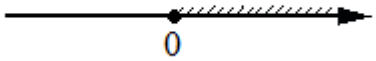
2) 

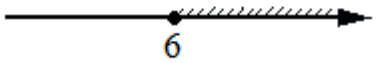
3) 

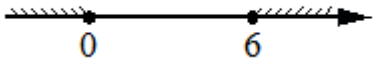
4) 

4. Укажите решение неравенства $6x - x^2 \leq 0$.

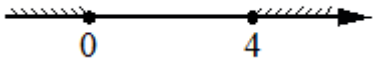
1) 

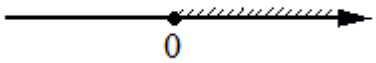
2) 

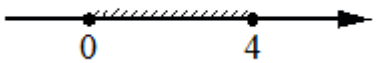
3) 

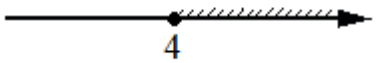
4) 

5. Укажите решение неравенства $4x - x^2 \leq 0$.

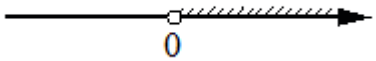
1) 

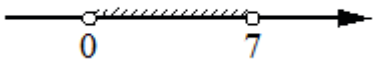
2) 

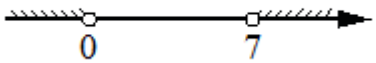
3) 

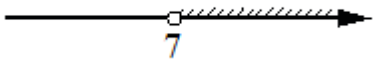
4) 

6. Укажите решение неравенства $7x - x^2 > 0$.

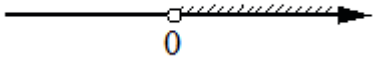
1) 

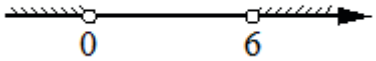
2) 

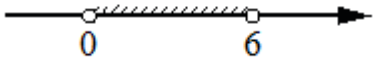
3) 

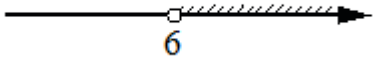
4) 

7. Укажите решение неравенства $6x - x^2 > 0$.

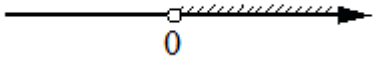
1) 

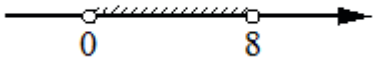
2) 

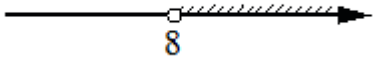
3) 

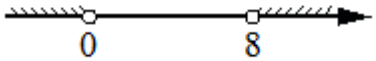
4) 

8. Укажите решение неравенства $8x - x^2 > 0$.

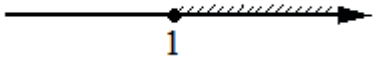
1) 

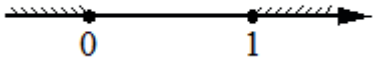
2) 

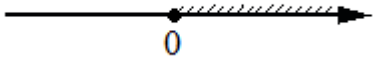
3) 

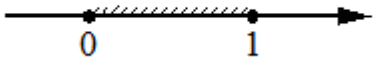
4) 

9. Укажите решение неравенства $x - x^2 \geq 0$.

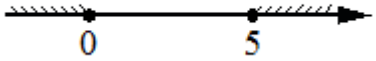
1) 

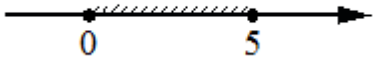
2) 

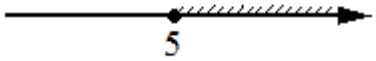
3) 

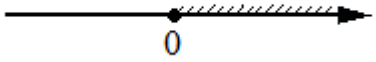
4) 

10. Укажите решение неравенства $5x - x^2 \geq 0$.

1) 

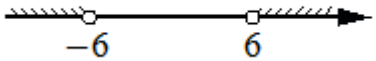
2) 

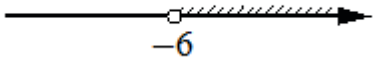
3) 

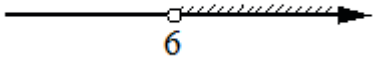
4) 

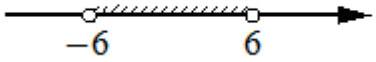
Задача 19

1. Укажите решение неравенства $x^2 > 36$.

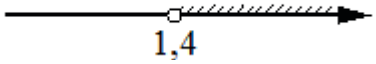
1) 

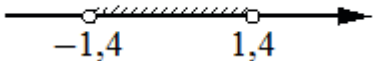
2) 

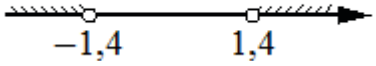
3) 

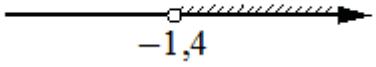
4) 

2. Укажите решение неравенства $25x^2 > 49$.

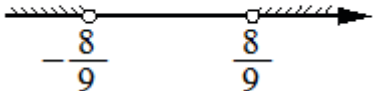
1) 

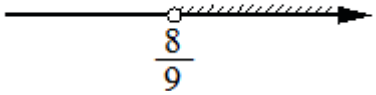
2) 

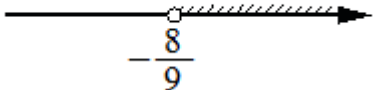
3) 

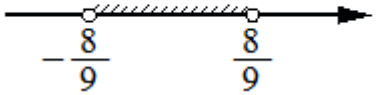
4) 

3. Укажите решение неравенства $81x^2 > 64$.

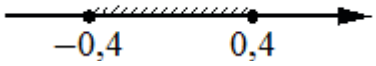
1) 

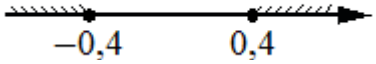
2) 

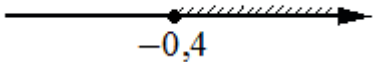
3) 

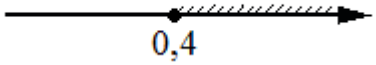
4) 

4. Укажите решение неравенства $25x^2 \geq 4$.

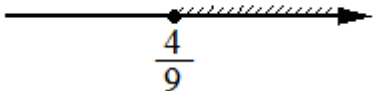
1) 

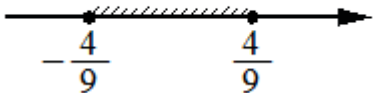
2) 

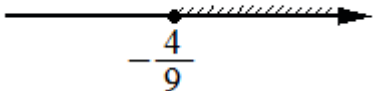
3) 

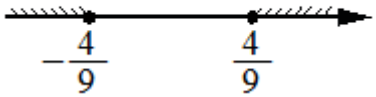
4) 

5. Укажите решение неравенства $81x^2 \geq 16$.

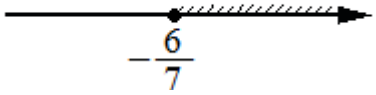
1) 

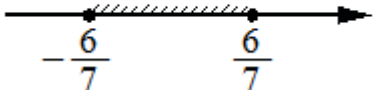
2) 

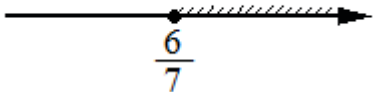
3) 

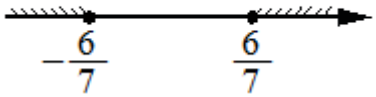
4) 

6. Укажите решение неравенства $49x^2 \geq 36$.

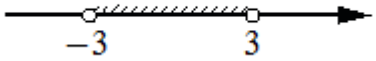
1) 

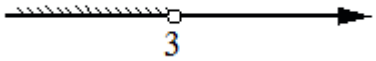
2) 

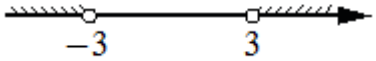
3) 

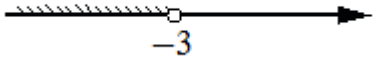
4) 

7. Укажите решение неравенства $x^2 < 9$.

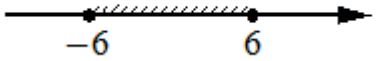
1) 

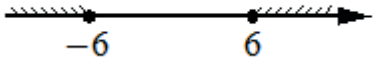
2) 

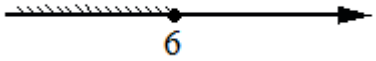
3) 

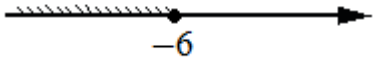
4) 

8. Укажите решение неравенства $x^2 \leq 36$.

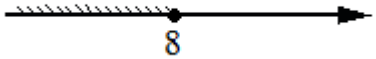
1) 

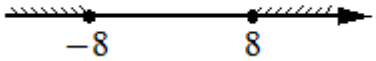
2) 

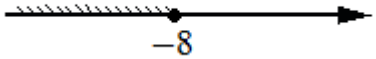
3) 

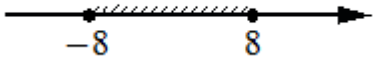
4) 

9. Укажите решение неравенства $x^2 \leq 64$.

1) 

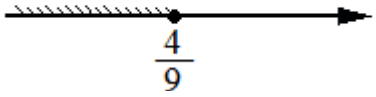
2) 

3) 

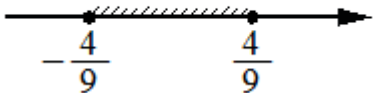
4) 

10. Укажите решение неравенства $81x^2 \leq 16$.

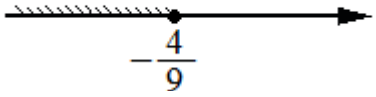
1)



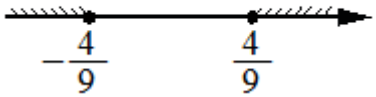
2)



3)



4)



Ответы

Задача 1

1. 1
2. 2
3. 2
4. 1
5. 4
6. 3
7. 2
8. 2
9. 2
10. 3

Задача 2

1. 2
2. 1
3. 1
4. 2
5. 1
6. 2
7. 3
8. 1
9. 2
10. 2

Задача 3

1. 2
2. 1
3. 3
4. 4
5. 1
6. 4
7. 3
8. 2
9. 4
10. 1

Задача 4

1. 3
2. 3
3. 3
4. 1
5. 1
6. 2
7. 3
8. 4
9. 2
10. 2

Задача 5

1. 3
2. 2
3. 4
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7. 3
8. 4
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10. 4

Задача 6

1. 1
2. 4
3. 2
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8. 4
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10. 3

Задача 7

1. 1
2. 4
3. 2
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8. 4
9. 2
10. 3

Задача 8

1. 2
2. 3
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7. 4
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9. 1
10. 4

Задача 9

1. 4
2. 3
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7. 3
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9. 4
10. 1

Задача 10

1. 1
2. 2
3. 3
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5. 1
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7. 3
8. 4
9. 1
10. 2

Задача 11

1. 1
2. 2
3. 3
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5. 1
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7. 3
8. 4
9. 1
10. 3

Задача 12

1. 1
2. 2
3. 3
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5. 1
6. 4
7. 3
8. 2
9. 4
10. 2

Задача 13

1. 3
2. 3
3. 4
4. 2
5. 1
6. 3
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8. 3
9. 3
10. 2

Задача 14

1. 1
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3. 1
4. 1
5. 2
6. 3
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8. 4
9. 2
10. 1

Задача 15

1. 1
2. 1
3. 3
4. 3
5. 3
6. 3
7. 1
8. 2
9. 2
10. 1

Задача 16

1. 2
2. 1
3. 1
4. 4
5. 2
6. 4
7. 3
8. 1
9. 3
10. 2

Задача 17

1. 3
2. 2
3. 4
4. 2
5. 3
6. 2
7. 2
8. 1
9. 4
10. 1

Задача 18

1. 4
2. 1
3. 3
4. 1
5. 3
6. 2
7. 3
8. 2
9. 4
10. 2

Задача 19

1. 1
2. 3
3. 1
4. 2
5. 4
6. 4
7. 1
8. 1
9. 4
10. 2